EXERCISE 1

(I) $E[(x_1+x_2)^2] = E[x_1^2 + x_2^2 + 2x_1x_2] =$ linearity + independence

$= \underbrace{E[x_1^2]}_{5} + \underbrace{E[x_2^2]}_{5} + \underbrace{2E[x_1]E[x_2]}_{2 \cdot 1 \cdot 1 \,\to\, 2} = 12$

$Var(X) = E[X^2] - E^2[X]$ $\quad (E[X])^2$

A

$E[x_1^2] = Var(x_1) + E^2[x_1] = 4 + 1 = 5$

(II) $E[x_1] = \frac{1}{2} \cdot 0 + \frac{1}{2} \cdot 2 = 1$

$Var(x_1) = \underbrace{E[x_1^2]}_{=2} - E^2[x_1] = 2 - 1 = 1$

$E[x_1(x_2+1)] = E(x_1)E[x_2+1] = 1 \cdot 2$

otherwise

$E[x_1x_2 + x_1] = E[x_1x_2] + E[x_1] = 2$ $\Rightarrow$ B, D are incorrect

$Y := \min\{x_1, x_2\}$ $\qquad Y \in \{0, 2\}$

$P(Y=2) = P(x_1 = 2 \text{ and } x_2 = 2) = 1/4$

$P(Y=0) = 3/4$

$E[Y] = 2 \cdot \frac{1}{4} + 0 \cdot \frac{3}{4} = 1/2$

A

$X \sim Exp(b)$ $\quad E[X] = \frac{1}{b}$ $\quad Var(X) = \frac{1}{b^2}$ $\quad X \sim Exp(2)$

$Y \sim Poisson(\mu)$ $\quad E[Y] = \mu$ $\quad Var(Y) = \mu$ $\quad Y \sim Poisson(2)$

ind.

$$Var(4X + 3Y - 1) = Var(4X) + Var(3Y) + Var(-1)$$

$$= 16 \cdot \frac{1}{4} + 9 \cdot 2 = 4 + 18 = 22$$

$\Rightarrow$ B, D not correct

$$E[4X^2 + 3Y^2] = 4E[X^2] + 3E[Y^2]$$

$$= 4 \cdot \frac{1}{2} + 3 \cdot 6 = 2 + 18 = 20$$

A

$E[X^2] = Var(X) + E^2[X]$

$= \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$

$E[Y^2] = Var(Y) + E^2[Y]$

$= 2 + 4 = 6$

EXERCISE 2

$X \in \{1, \dots, 6\}$

FAIR DIE \ RESULT	1	2	3	4	5	6
YES	$\frac{1}{2} \cdot \frac{1}{6}$	$\frac{1}{2} \cdot \frac{1}{6}$	$\frac{1}{2} \cdot \frac{1}{6}$	$\frac{1}{2} \cdot \frac{1}{6}$	$\frac{1}{2} \cdot \frac{1}{6}$	$\frac{1}{2} \cdot \frac{1}{6}$
NO	$\frac{1}{2} \cdot \frac{1}{10}$	$\frac{1}{2} \cdot \frac{1}{2}$	$\frac{1}{2} \frac{1}{10}$	$\frac{1}{2} \frac{1}{10}$	$\frac{1}{2} \frac{1}{10}$	$\frac{1}{2} \frac{1}{10}$
X :	$2/15$	$1/3$	$2/15$	$2/15$	$2/15$	$2/15$

marginal pmf of X, namely $P_X(x)$

$E[X] = 3.2$

$Var(X) = E[X^2] - E^2[X] = 12.933 - 10.24 = 2.693$

ESERCIZIO 3

- $E[\bar{x}] = E\left[\frac{1}{n}\sum x_i\right] = \frac{1}{n}\underbrace{\sum \overbrace{E[x_i]}^{\mu}}_{n\mu} = \mu$

- $Var(\bar{x}) = Var\left(\frac{1}{n}\sum x_i\right) \underset{ind}{=} \frac{1}{n^2}\sum Var(x_i) = \frac{\sigma^2}{n}$

- $E[\bar{x} + x_1] = \mu + \mu = 2\mu$ $\quad (= E[\bar{x}] + E[x_1])$

- $Var(\bar{x} + x_1) \cancel{=} \frac{\sigma^2}{n} + \sigma^2$ NOOOOOO

$x_1, \bar{x}$ are NOT INDEP.!

$$Var\left(\frac{1}{n}\sum_i x_i + x_1\right) = Var\left(\frac{1}{n}\sum_{i\neq 1} x_i + \frac{1}{n}x_1 + x_1\right) =$$

$$= Var\left(\frac{1}{n}\sum_{i\neq 1} x_i + \left(1 + \frac{1}{n}\right)x_1\right) =$$

$$= \frac{1}{n^2}\underbrace{Var\left(\sum_{i\neq 1} x_i\right)}_{(n-1)\sigma^2} + \left(1 + \frac{1}{n}\right)^2 \underbrace{Var(x_1)}_{\sigma^2} =$$

$$= \frac{1}{n}\sigma^2 - \cancel{\frac{\sigma^2}{n^2}} + \sigma^2 + \frac{2}{n}\sigma^2 + \cancel{\frac{1}{n^2}\sigma^2} = \sigma^2 + \frac{3}{n}\sigma^2$$

EXERCISE 4

(I) X is sum of 3 bernoulli by definition

Y is sum of 7 bernoulli by definition

they are all independent!

$\Rightarrow X+Y$ is the sum of 10 i.i.d. Bernoulli (0.2)

$\Rightarrow X+Y \sim \text{Binomial}(10, 0.2)$

$\Rightarrow P(X+Y=3) = \binom{10}{3} 0.2^3 \; 0.8^7$ $\boxed{C}$

(II) Note that $XY \sim \text{Bernoulli}(2/9)$ as

$XY \in \{0,1\}$ and $P(XY=1) = P(X=1, Y=1) = 2/9$

$E[XY] = 2/9 \quad P(XY=0) = \frac{7}{9} \quad P(XY=1) = \frac{2}{9}$ $\boxed{C}$

EXERCICE 4

III

$$P(2XY > -1) = P(Y=0, \overbrace{2X\cdot 0 > -1}) + P(Y=1, \underbrace{2X\cdot 1 > -1}) =$$

$$= \underbrace{P(Y=0)}_{2/3}\underbrace{P(\overbrace{2X\cdot 0}^{0} > -1)}_{1} + \underbrace{P(Y=1)}_{1/3}\underbrace{P(2X > -1)}$$

$$= 0.9244$$

$$1 - \text{pnorm}(-1/2, 1, 2)$$

$$P(2XY > 1) = \underbrace{P(Y=0)\overbrace{P(2X\cdot 0 > 1)}^{=0}}_{0} + \underbrace{P(Y=1)}_{1/3}\underbrace{P(2X > 1)}$$

$$1 - \text{pnorm}(1/2, 1, 2)$$

$$= 0.1996$$

$$\boxed{B}$$

EXERCISE

$X \backslash Y$	0	1	P_X
-1	0	-1 $p = 2/6$	$1/3$
0	0 $p = 2/6$	0 0	$1/3$
1	0 $1/6$	1 $1/6$	$1/3$
P_Y	$1/2$	$1/2$	

U

(a) $p = 2/6$

(b) $P_X(-1) = 2/6 = P_X(0) = P_X(1)$

$P_Y(0) = P_Y(1) = 1/2$

They are NOT independent e.g. $P(X=0, Y=1) = 0$

$P(X=0)\,P(Y=1) = 1/6$

(c) $E[X] = \frac{1}{3} \cdot -1 + \frac{1}{3} \cdot 0 + \frac{1}{3} 1 = 0$

(d) $E[Y] = 1/2$

$Var(Y) = \Sigma (y - E[Y])^2 = (0 - 1/2)^2 \cdot 1/2 + (1 - 1/2)^2 \, 1/2$

$\hookrightarrow \underbrace{E[Y^2]}_{1/2} - \underbrace{E[Y]^2}_{1/2^2} = 1/2 - 1/2^2 = 1/4$

$\hookrightarrow 0^2 \cdot \frac{1}{2} + 1^2 \cdot \frac{1}{2} = 1/2$

(e) $U := X \cdot Y \qquad U \in \{0, 1, -1\}$

$P_U(0) = 1/2 \qquad P_U(1) = 1/6 \qquad P_U(-1) = 2/6$

$E[U] = 0 \cdot \frac{1}{2} + 1 \cdot 1/6 - 1 \cdot 2/6 = -1/6$

$Var(U) = E[U^2] - E[U]^2 = 1/2 - 1/36 = \ldots$

not required

$\hookrightarrow 0^2 \cdot \frac{1}{2} + 1^2 \cdot 1/6 + (-1)^2 \cdot \frac{2}{6} = 1/2$